Moein Heidari

✉ moeinheidari7829@gmail.com | LinkedIn | Google Scholar | GitHub | Website

Education

- **The University of British Columbia (UBC)** — Vancouver, BC, Canada
  Ph.D. Candidate – Biomedical Engineering (advanced to candidacy Nov 2025) — *Jan 2024 – 2028*
  Thesis: Towards Complete AI-Enabled Echocardiography Interpretation with a Unified Multimodal Framework
  Supervisor: Dr. Ilker Hacihaliloglu

- **Iran University of Science and Technology (IUST)** — Tehran, Iran
  M.Sc. – Communication Systems; GPA: 4.0 (18.48/20) — *Oct 2021 – Nov 2023*
  Thesis: Fully Transformer-based End-to-End Communication System (Mark: Very Good)
  Supervisor: Dr. Shahrokh Farahmand

- **Iran University of Science and Technology (IUST)** — Tehran, Iran
  B.Sc. – Electrical Eng. – Communications; GPA: 3.6 (17.11/20) — *Sep 2017 – Sep 2021*
  Thesis: Deep Learning Based End-to-End Wireless Communication System With Conditional GAN as Unknown Channel (19.5/20)
  Supervisor: Dr. Shahrokh Farahmand

Publications

[1] †M. Heidari, J. Rao, J. Choraria, W. Chen, D. J. Foran, I. Hacihaliloglu, "Which Pretext Task Transfers? Self-Supervised Pretraining Objectives for Lung Ultrasound," SPIE Medical Imaging 2027, **Oral**, arXiv, GitHub.

[2] †M. Heidari, J. Rao, Y. Chen, Y. Medghalchi, R. Jiang, W. Chen, D. J. Foran, J. Grewal, I. Hacihaliloglu, "Every Pixel Is an Option: A System One Model for Medical Ultrasound Biometry," arXiv preprint, 2026, Project, arXiv.

[3] M. Yazdani, P. Ashrafian, M. Heidari, A. Rafiei Sardouei, D. Shahriari, I. Hacihaliloglu, "Incorporating Synthetic Data from Diffusion Models Supports Echocardiography Downstream Tasks," Journal of Medical Imaging (JMI), 2026, Paper.

[4] †M. Heidari, Y. Huang, Y. Medghalchi, A. Rafiei, R. Tam, I. Hacihaliloglu, "WaRA: Wavelet Low-Rank Adaptation for Medical Image Classification," MICCAI 2026 MIRASOL Workshop, **Oral**, **Impact Award**, Project, arXiv, GitHub.

[5] †M. Heidari, A. Mehrabian, M. A. Roohi, R. Jiang, W. Chen, D. J. Foran, J. Grewal, I. Hacihaliloglu, "Echo-CoPilot: A Multiple-Perspective Agentic Framework for Reliable Echocardiography Interpretation," MICCAI 2026 Agentic AI for Medicine Workshop, Project, arXiv, GitHub.

[6] A. Kimbowa, †M. Heidari, D. Liu, I. Hacihaliloglu, "XTinyU-Net: Training-Free U-Net Scaling via Initialization-Time Sensitivity," **MICCAI 2026 (early accept)**, arXiv, GitHub.

[7] A. Bozorgpour, S. Ghorbani Kolahi, M. Heidari, I. Hacihaliloglu, D. Merhof, "MaLViL: Multi-axis Low-rank Vision-LSTM for Medical Image Segmentation," MICCAI 2026 MLMI Workshop, arXiv, GitHub.

[8] †M. Heidari, A. Bozorgpour, A. Zarif-Fakharnia, W. Chen, D. Merhof, D. J. Foran, J. Grewal, I. Hacihaliloglu, "Echo-E^{3}Net: Efficient Endocardial Spatio-Temporal Network for Ejection Fraction Estimation," MICCAI 2026 ASMUS Workshop, Project, arXiv, GitHub.

[9] Y. Medghalchi, M. Heidari, C. Allard, L. Sigal, I. Hacihaliloglu, "Prompt2Perturb (P2P): Text-Guided Diffusion-Based Adversarial Attacks on Breast Ultrasound Images," **CVPR 2025**, arXiv, GitHub.

[10] †M. Heidari, R. Rezaeian, R. Azad, D. Merhof, H. Soltanian-Zadeh, I. Hacihaliloglu, "SL^{2}A-INR: Single-Layer Learnable Activation for Implicit Neural Representation," **ICCV 2025**, Project, arXiv, GitHub, Video.

[11] A. Mehrabian, P. Mojarad Adi, M. Heidari, I. Hacihaliloglu, "Implicit Neural Representations with Fourier Kolmogorov-Arnold Networks," ICASSP 2025, arXiv, GitHub.

[12] †M. Heidari, E. Khodapanah Aghdam, A. Manzella, D. Hsu, R. Scalabrino, W. Chen, D. J. Foran, I. Hacihaliloglu, "A Study on the Performance of U-Net Modifications in Retroperitoneal Tumor Segmentation," SPIE Medical Imaging 2025, Project, arXiv, GitHub.

[13] S. Ghorbani Kolahi, S. K. Chaharsooghi, T. Khatibi, A. Bozorgpour, R. Azad, M. Heidari, I. Hacihaliloglu, D. Merhof, "MSA^{2}Net: Multi-scale Adaptive Attention-guided Network for Medical Image Segmentation," BMVC 2024, arXiv, GitHub.

[14] †M. Heidari, S. Ghorbani Kolahi, S. Karimijafarbigloo, B. Azad, A. Bozorgpour, S. Hatami, R. Azad, A. Diba, U. Bagci, D. Merhof, "Computation-Efficient Era: A Comprehensive Survey of State Space Models in Medical Image Analysis," arXiv preprint, 2024, Project, arXiv, GitHub.

[15] †M. Heidari, R. Azad, S. Ghorbani Kolahi, R. Arimond, L. Niggemeier, A. Sulaiman, A. Bozorgpour, E. Khodapanah Aghdam, A. Kazerouni, I. Hacihaliloglu, D. Merhof, "Enhancing Efficiency in Vision Transformer Networks: Design Techniques and Insights," arXiv preprint, 2024, Project, arXiv.

[16] P. Ashrafian, M. Yazdani, M. Heidari, D. Shahriari, I. Hacihaliloglu, "Vision-Language Synthetic Data Enhances Echocardiography Downstream Tasks," arXiv preprint, 2024, arXiv, GitHub.

[17] R. Azad, A. Kazerouni, M. Heidari, E. Khodapanah Aghdam, A. Molaei, Y. Jia, A. Jose, R. Roy, D. Merhof, "Advances in Medical Image Analysis with Vision Transformers: A Comprehensive Review," Medical Image Analysis (MedIA), 2024, arXiv, GitHub.

[18] M. Fiaz, †M. Heidari, R. M. Anwer, H. Cholakkal, "SA^2-Net: Scale-aware Attention Network for Microscopic Image Segmentation," BMVC 2023, **Oral**, arXiv, GitHub.

[19] †M. Heidari, A. Kazerouni, M. Soltany, R. Azad, E. Khodapanah Aghdam, J. Cohen-Adad, D. Merhof, "HiFormer: Hierarchical Multi-scale Representations Using Transformers for Medical Image Segmentation," WACV 2023, Project, arXiv, GitHub.

[20] A. Kazerouni, E. Khodapanah Aghdam, M. Heidari, R. Azad, M. Fayyaz, I. Hacihaliloglu, D. Merhof, "Diffusion Models in Medical Imaging: A Comprehensive Survey," Medical Image Analysis (MedIA), 2023, arXiv, GitHub.

[21] †M. Heidari, A. Morsali, T. Abedini, S. Heydarian, "DiffGANPaint: Fast Inpainting Using Denoising Diffusion GANs," ICLR 2023 Tiny Papers, Project, arXiv.

[22] A. Morsali, M. Heidari, S. Heydarian, T. Abedini, "MLP-Attention: Improving Transformer Architecture with MLP Attention Weights," ICLR 2023 Tiny Papers, Paper, GitHub.

[23] R. Azad, M. Heidari, K. Yilmaz, M. Hüttemann, S. Karimijafarbigloo, Y. Wu, A. Schmeink, D. Merhof, "Loss Functions in the Era of Semantic Segmentation: A Survey and Outlook," arXiv preprint, 2023, arXiv, GitHub.

[24] R. Azad, M. Heidari, Y. Wu, D. Merhof, "Contextual Attention Network: Transformer Meets U-Net," MICCAI 2022 MLMI Workshop, arXiv.

[25] R. Azad, M. Heidari, J. Cohen-Adad, E. Adeli, D. Merhof, "Intervertebral Disc Labeling With Learning Shape Information, A Look Once Approach," MICCAI 2022 PRIME Workshop, arXiv.

[26] R. Azad, M. Heidari, M. Shariatnia, E. Khodapanah Aghdam, S. Karimijafarbigloo, E. Adeli, D. Merhof, "TransDeepLab: Convolution-Free Transformer-based DeepLab v3+ for Medical Image Segmentation," MICCAI 2022 PRIME Workshop, arXiv.

[27] R. Azad, M. T. AL-Antary, M. Heidari, D. Merhof, "TransNorm: Transformer Provides a Strong Spatial Normalization Mechanism for a Deep Segmentation Model," IEEE Access, 2022, Paper.

† *First / co-first author*

Research Interests

Agentic AI & LLM Agents · Multimodal LLMs & Vision-Language Models · Diffusion & Generative Models · Self-Supervised Learning · Foundation Models & Parameter-Efficient Fine-Tuning · Implicit Neural Representations · Efficient Architectures (State Space Models, Lightweight U-Nets) · Computer Vision · Medical Image Analysis (Ultrasound & Echocardiography)

Skills Summary

- **Languages**: Python, MATLAB, C/C++, SQL, Julia
- **Frameworks**: PyTorch, TensorFlow, Keras, Flux, OpenCV, NumPy, Pandas, Matplotlib
- **Tools**: Linux, LaTeX, Git, MySQL
- **Soft Skills**: Leadership, Event Management, Writing, Public Speaking, Time Management

Research Experience

- **Mohamed bin Zayed University of Artificial Intelligence** — Remote
  Research Assistant (Supervisor: Dr. Hisham Cholakkal) — *Jan 2023 – May 2023*
  - **Attention Mechanisms**: Research on attention mechanisms for microscopic medical image segmentation, resulting in 1 conference publication.
- **RWTH Aachen University** — Remote
  Research Assistant (Supervisors: Prof. Dorit Merhof and Reza Azad) — *Oct 2021 – Dec 2023*
  - **Transformer Models**: Research on Transformer models resulting in 4 conference and 3 journal papers.
  - **Intervertebral Disc Labeling**: Segmentation of intervertebral discs from medical images, resulting in 1 conference publication.
- **Iran University of Science and Technology (Dept. of CS)** — Tehran, Iran
  Research Assistant (Supervisor: Dr. Mohammad Reza Mohammadi) — *Apr 2021 – Jul 2021*
  - **Self-Supervised Object Detection**: Trained RetinaNet with various self-supervised pretraining methods (MOCO, PIRL, etc.) for object detection with low supervision.
- **DGSculptor, Montreal, Canada** — Remote
  Machine Learning and Computer Vision Researcher — *Dec 2021 – Dec 2023*
  - **Generative Flow Networks**: Research on various generative models and their statistical perspective, resulting in 2 conference publications.

Teaching Assistant Experience

- **DSCI_V 430 – Fairness, Accountability, Transparency and Ethics (FATE) in DS** — UBC, Canada
  Teaching Assistant — *Winter 2025 (2025W2)*
- **CPSC 330 – Applied Machine Learning** — UBC, Canada
  Teaching Assistant — *Winter 2024 (2024W2)*
- **DSCI 100 – Introduction to Data Science** — UBC, Canada
  Teaching Assistant — *Fall 2025 (2025W2)*
- **CPSC 340 – Machine Learning and Data Mining** — UBC, Canada
  Teaching Assistant — *Winter 2024 (2024W2)*
- **Signals and Systems Analysis** — IUST, Iran
  Teaching Assistant — *Spring 2021, Spring 2022, Spring 2023*
- **Fundamentals of Deep Learning** — IUST, Iran
  Teaching Assistant — *Spring 2022*

Working Experience

- **AI Engineer** — Tehran, Iran
  FaraAI (faraai.ir) — *May 2022 – Dec 2022*
  - **Responsibilities**: Designed and implemented computer vision pipelines for automatic detection and dispatching system for roadside emergencies, including fires and traffic incidents.

Talks

- **Research Proposal (Ph.D. Candidacy)** — UBC, Canada
  Doctoral committee presentation — *Nov 2025*
  Towards Complete AI-Enabled Echocardiography Interpretation with a Unified Multimodal Framework
- **Research Presentation** — UBC, Canada
  Annual HLI Research Day — *Aug 2025*

Professional Services

NeurIPS	Reviewer, 2026
MICCAI	Reviewer, 2024, 2025, 2026 *(Rank 1 in AI for medical imaging)*
CVPR	Reviewer, 2025 *(Rank 1 in Computer Vision)*
ICCV	Reviewer, 2025 *(Rank 2 in Computer Vision)*
ICLR	Reviewer, 2024, 2025, 2026 *(Rank 1 in Machine Learning)*
ICML	Reviewer, 2026
BMVC	Reviewer, 2025, 2026
TMLR	Reviewer
IEEE TMI	Journal Reviewer *(Top-tier journal in medical imaging)*
IEEE JBHI	Journal Reviewer *(Top-ranked journal for biomedical AI)*
IEEE Access	Journal Reviewer
IEEE Sys. J.	Journal Reviewer
IEEE SPL	Journal Reviewer

Relevant Course Grades

- Probability & Statistics: 19.25/20 (IUST)
- Digital Image Processing: 17.00/20 (IUST)
- Deep Learning: 20.0/20 (IUST)
- Advanced Data Mining: 20.0/20 (IUST)
- Cellular Communication: 19.25/20 (IUST)
- Engineering Mathematics: 20.0/20 (IUST)
- Reinforcement Learning: 20.0/20 (IUST)
- Random Processes: 19.25/20 (IUST)
- BMEG 591 – Topics in Biomed. Eng.: 92/100 (UBC)
- BMEG 581 – Prof. & Acad. Dev.: 100/100 (UBC)

Honors and Awards

- **Impact Award**, MIRASOL Workshop at MICCAI 2026, for "WaRA: Wavelet Low-Rank Adaptation for Medical Image Classification" – Oct 2026
- **ICCV 2025 Broadening Participation Program award** to present at ICCV 2025, Honolulu – Sep 2025
- Ranked **3rd** among 25 Communications students, IUST, Iran – May 2021
- Ranked **9th** among 127 Electrical Engineering students, IUST, Iran – May 2021
- Ranked **1st** in the national Rahneshan competition for detecting inappropriate content in images/videos, INEF – Feb 2021
- Top 1% among ∼148,000 participants in the National University Entrance Exam, Iran – Summer 2017
- Received honorary master's admission from Iran University of Science and Technology – Sep 2021
- Top Student for 5 consecutive semesters at the Iran Languages Institute (ILI) – 2015–2016

Online Courses

- Convolutional Neural Networks, Coursera
- Structuring ML Projects, Coursera
- GANs Specialization, Coursera
- Neural Networks and Deep Learning, Coursera
- Sequence Models, Coursera
- Improving Deep NNs: Hyperparameter Tuning, Regularization & Optimization, Coursera

Languages

- IELTS (Academic): Listening 7.5, Reading 7, Speaking 7.5, Writing 6.5, **Overall: 7** – C1 Proficiency
- Persian: Native

References

Dr. Ilker Hacihaliloglu
Assistant Professor, UBC
ilker.hacihaliloglu@ubc.ca

Prof. David J. Foran
Prof. of Pathology, Lab Medicine & Radiology
Rutgers Robert Wood Johnson Medical School
foran@cinj.rutgers.edu

Prof. Dorit Merhof
Prof., Informatics & Computer Science
University of Regensburg
dorit.merhof@lfb.rwth-aachen.de